Data Mining HW3

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Q: Explain your implementation which get the best performance in detail.

- Data Preparation:

  I first loaded the data using pandas, ensuring a clear separation of features (X) and labels (y). To maintain a balanced distribution of labels across training and validation sets, I applied stratified sampling during the train-validation split. This approach helps in preserving the proportion of each class, which is essential for training unbiased models.

- Standardization:

  I standardized the feature data using a StandardScaler. This step is crucial as it normalizes the features, preventing any single feature with a large scale from disproportionately influencing the model's learning process. Standardization ensures that the model treats all features equally, which is particularly important for algorithms like RandomForest.

- Model Training:

  I employed a RandomForestClassifier configured with 30 trees and used 'entropy' as the criterion for the best splits based on information gain. RandomForest, being an ensemble method, combines the predictions of several base estimators built with a given learning algorithm to improve generalizability and robustness against overfitting.

- Validation:

  I calculated the ROC AUC score. This metric is used to evaluate the model's ability to differentiate between the classes. A higher ROC AUC value indicates that the model can effectively distinguish between positive and negative classes, which is crucial for classification tasks.

- Testing:

  During the testing phase, I predicted probabilities for the test set and calculated the Mean Squared Error (MSE) against a uniform distribution. This MSE was then inverted to derive a score, with higher values indicating better performance.

Q: Explain the rationale for using auc score instead of F1 score for binary classification in this homework.

1. Imbalanced Classes

   The AUC score is particularly useful for datasets with imbalanced classes, like ours, where outliers might be much fewer than normal instances. It evaluates the model across all classification thresholds, thereby providing a measure of the model's ability to distinguish between the classes, independent of any imbalance.

2. Threshold Flexibility

   AUC evaluates the model's ability to correctly rank predictions rather than focusing solely on a specific threshold. This measure ensures performance consistency across various threshold settings, which is critical in anomaly detection where defining an exact threshold for outlier classification isn't straightforward.

3. Multi-class Adaptation

   Although AUC is primarily designed for binary classification, it can also be applied to multi-class classification situations by comparing each class to all others. Thus, AUC is a generalized metric for problems where anomalies are not strictly binary categorized.

Q: Discuss the difference between semi-supervised learning and unsupervised learning.

- Definition and Usage:
  - Semi-Supervised Learning

    In this anomaly detection task, semi-supervised learning involves combining a small amount of labeled data with a larger amount of unlabeled data. This approach allows the model to learn from both types of data, enhancing its ability to detect anomalies even when the test set lacks labels.
  - Unsupervised Learning

    On the other hand, unsupervised learning relies entirely on the data's inherent structure without using any labels. This method is beneficial when the test data lacks labels, requiring the model to identify patterns and outliers based solely on data features.
- Model Training and Handling Outliers:
  - Semi-Supervised Learning

    Utilizing both labeled and unlabeled data improves the model's sensitivity to anomalies, especially in distinguishing normal from abnormal patterns, as the model gets some guidance from the labeled data. Including labeled examples during training helps the model identify features that indicate anomalies, reducing false positives and improving accuracy.
  - Unsupervised Learning

    The model uses only data features to learn patterns and detect anomalies, providing valuable insights into the data's structure. However, detecting outliers based on deviations can be less effective

when outliers do not conform to clear patterns, requiring careful tuning to avoid misclassification.